Ben M. Andrew

benjamin.andrew@manchester.ac.uk | benmandrew.com

Education

University of Manchester, PhD in Computer Science Sep 2024 – Present

- Research focused on formal verification of degraded systems.
- Supervised by Dr. Marie Farrell, Prof. Louise Dennis, and Prof. Michael Fisher.

University of Cambridge, BA (Hons) in Computer Science Oct 2019 – Jun 2022

- Grade II.1 (70%).
- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing it against traditional protocols. Supervised by Dr. Jackson Woodruff.

Publications

- Andrew, B. M., Dennis, L. A., Fisher, M., Farrell, M. *Counterexample-Guided Interval Weakening*. Rigorous State-Based Methods (ABZ) 2026. 10.1007/978-3-032-26752-8_1
- Andrew, B. M. *Weakening Goals in Logical Specifications*. Rigorous State-Based Methods (ABZ) 2025. 10.1007/978-3-031-94533-5_22

Skills

Programming Languages: OCaml, C++, C, Python, Rust

Tools: Git, Docker, Claude Code

Spoken Languages: English, French

Industry Experience

Software Engineer, Tarides – Paris, France Sep 2022 – Jun 2024

- Built open-source CI infrastructure for the OCaml ecosystem, distributing dependency solving, compilation, and testing across a broad matrix of operating systems, CPU architectures, and compiler versions.
- Tested tens of thousands of packages across the Opam Repository, OCaml's central package universe.
- Exposed concrete bugs in the OCaml 5 multicore runtime through systematic stress-testing, reporting and tracking them to resolution upstream.

Embedded Systems Intern, Tarides – Paris, France Jul – Sep 2021

- Cross-compiled the OCaml bytecode runtime to an ARM-based microcontroller running a real-time operating system, as part of an R&D project.

Computer Graphics Research Intern, University of Cambridge Jun – Sep 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Teaching Experience

University of Cambridge, Supervisor (TA equiv.) Oct 2022 – Present

- Leading small group supervisions for undergraduate courses in computer science, including topics such as logic, formal verification, and semantics of programming languages.

University of Manchester, Graduate Teaching Assistant Oct 2024 – May 2026

- Assisted in teaching graduate and undergraduate courses in computer science, including topics such as software engineering, logic, and formal verification.